Cheat sheet

Transformers & LLMs

Architecture, training, inference and model selection

Architecture

Attention	$\text{softmax}(QK^T/\sqrt{d_k})\ V$
Multi-head	Several attention heads concatenated
Causal mask	Decoder tokens see only previous tokens
Positional info	Sinusoidal, learned, RoPE (rotary), ALiBi
Block	Attention → add & norm → feed-forward → add & norm
Families	Encoder (BERT), decoder (GPT, Claude, Llama, Gemini), encoder-decoder (T5)
MoE	Mixture-of-Experts routes each token to a few expert networks; more capacity per FLOP

Training

Pretraining	Next-token prediction on trillions of tokens
SFT	Supervised instruction tuning
Preference tuning	RLHF (reward model + PPO), DPO, RLAIF / constitutional methods
Reasoning training	RL on verifiable tasks (maths, code) to improve step-by-step thinking
Scaling laws	Loss falls predictably with parameters, data and compute

Inference

Decoding	Greedy, beam search, sampling (temperature, top-p, top-k)
KV cache	Stores past keys/values to avoid recomputation
Quantisation	8-bit / 4-bit weights (GPTQ, AWQ, GGUF)
Speculative decoding	Small draft model proposes tokens; big model verifies

Batching	Continuous batching (e.g. vLLM) for throughput
Context length	Long context vs retrieval trade-off; "lost in the middle" effect

Choosing a model

Quality	Task benchmarks + your own evaluation set
Cost / latency	Price per million tokens, time-to-first-token, tokens per second
Privacy	API vs self-hosted open weights
Features	Tool calling, structured output, vision, long context, fine-tuning support
Licence	Check open-weight licences for commercial use